

Conformal Prediction for Hierarchical Data

Guillaume Principato

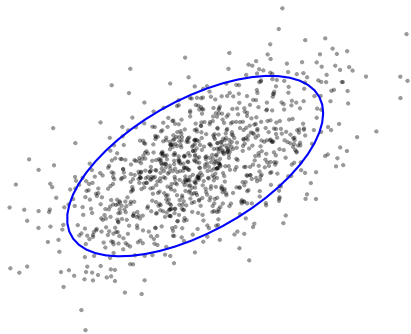
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Joint work with Gilles Stoltz, Yvenn Amara-Ouali, Yannig Goude,
Bachir Hamrouche and Jean-Michel Poggi



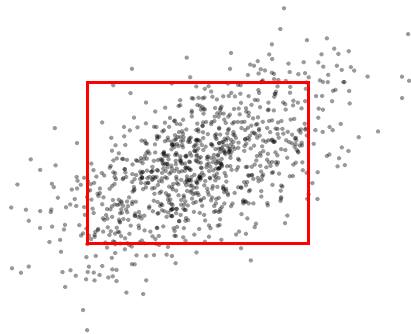
Prediction Regions for Multivariate Data: Validity

In regression for multivariate targets $\mathbf{y} \in \mathbb{R}^m$ and covariates $\mathbf{x} \in \mathbb{R}^d$ with $m > 1$, we can consider 2 types of prediction regions:



which targets joint coverage:

$$\mathbb{P}(\mathbf{y}_{T+1} \in C(\mathbf{x}_{T+1})) \approx 1 - \alpha$$

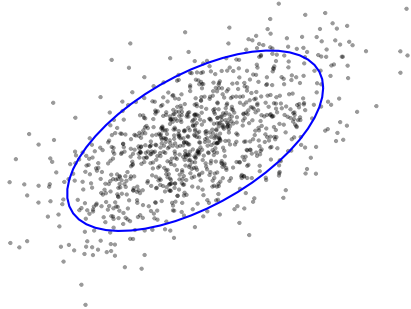


which targets component-wise coverage:

$$\forall i \in [m], \quad \mathbb{P}(y_{T+1,i} \in C_i(\mathbf{x}_{T+1})) \approx 1 - \alpha$$

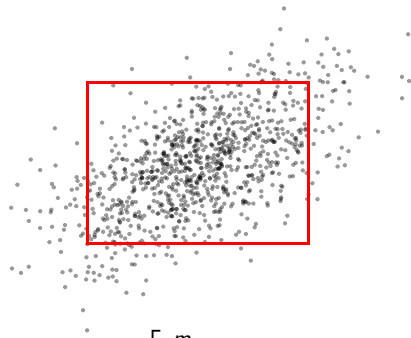
Prediction Regions for Multivariate Data: Efficiency

Since the coverage objective differs, the efficiency objectives of the prediction regions must be different:



minimizing $\mathbb{E}[\mathfrak{L}_m(C(\mathbf{x}_{T+1}))]$,

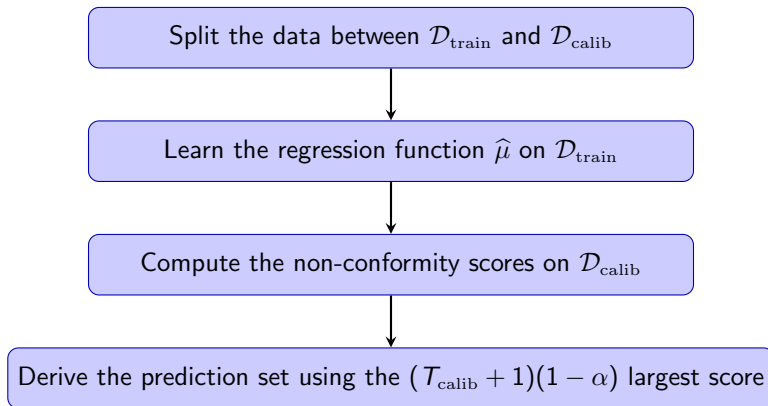
$\mathfrak{L}_m$ is the Lebesgue measure over $\mathbb{R}^m$



minimizing $\mathbb{E}\left[\sum_{i=1}^m w_i \mathfrak{L}_1(C_i(\mathbf{x}_{T+1}))^2\right]$,

$\mathbf{w} = (w_1, \dots, w_m)^\top$ a positive weight vector and $\mathfrak{L}_1$ the Lebesgue measure over $\mathbb{R}$

General principle of Split Conformal Prediction (SCP)



Ellipsoidal SCP for joint coverage: Algorithm 1 (baseline 1)

Split the data between $\mathcal{D}_{\text{train}}$, $\mathcal{D}_{\text{estim}}$ and $\mathcal{D}_{\text{calib}}$

$$\hat{\mu}(\cdot) = \mathcal{A}((\mathbf{x}_t, \mathbf{y}_t)_{t \in \mathcal{D}_{\text{train}}})$$

Compute a symmetric definite positive matrix $A = \mathcal{E}((\mathbf{x}_t, \mathbf{y}_t)_{t \in \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{estim}}})$

for $t \in \mathcal{D}_{\text{calib}}$, let $\hat{\mathbf{y}}_t = \hat{\mu}(\mathbf{x}_t)$ and compute the non-conformity score $\check{s}_t = \|\mathbf{y}_t - \hat{\mathbf{y}}_t\|_A$

Set $\check{E}(\cdot) = \left\{ \mathbf{y} \in \mathbb{R}^m : \|\mathbf{y} - \hat{\mu}(\cdot)\|_A \leq \check{s}_{(\lceil (\tau_{\text{calib}} + 1)(1 - \alpha) \rceil)} \right\}$ and return $\check{E}(\mathbf{x}_{T+1})$

Component-wise SCP: Algorithm 2 (baseline 2)

Split the data between $\mathcal{D}_{\text{train}}$, $\mathcal{D}_{\text{estim}}$ and $\mathcal{D}_{\text{calib}}$

$$\hat{\mu}(\cdot) = \mathcal{A}((\mathbf{x}_t, \mathbf{y}_t)_{t \in \mathcal{D}_{\text{train}}})$$

for $t \in \mathcal{D}_{\text{calib}}$, let $\hat{\mathbf{y}}_t = \hat{\mu}(\mathbf{x}_t)$ and compute the non-conformity score $\hat{\mathbf{s}}_t = \mathbf{y}_t - \hat{\mathbf{y}}_t$

for each $i \in [m]$, order the non-conformity scores: $\hat{s}_{(1),i} \leq \dots \leq \hat{s}_{(T_{\text{calib}}),i}$

Set and return $\hat{C}_i(\mathbf{x}_{T+1}) = \left[\hat{\mu}_i(\mathbf{x}_{T+1}) + \hat{s}_{(\lfloor (T_{\text{calib}}+1)\alpha/2 \rfloor),i}, \hat{\mu}_i(\mathbf{x}_{T+1}) + \hat{s}_{(\lceil (T_{\text{calib}}+1)(1-\alpha/2) \rceil),i} \right]$

Immediate Results for the baselines (informal)

By construction, if the $(\mathbf{x}_t, \mathbf{y}_t)_{t \in [T+1]}$ are i.i.d. (or exchangeable), the ellipsoidal prediction set output by Algorithm 1 achieves joint coverage:

$$\mathbb{P}(\mathbf{y}_{T+1} \in \check{E}(\mathbf{x}_{T+1})) \approx 1 - \alpha$$

Similarly, the prediction rectangle from Algorithm 2 achieves component-wise coverage:

$$\forall i \in [m], \quad \mathbb{P}(y_{T+1,i} \in \hat{C}_i(\mathbf{x}_{T+1})) \approx 1 - \alpha$$

Our Contributions

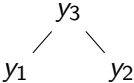
If the multivariate targets follow some hierarchical structure, adding a projection step to the aforementioned benchmarks Algorithms improves their efficiencies. Joint coverage admits a relatively simple analysis, but component-wise coverage raises deeper and more subtle theoretical difficulties.

Hierarchical Data and Forecast Reconciliation

Notations

Denote $\mathbf{y}_{1:n} \in \mathbb{R}^n$ the n first components of $\mathbf{y} \in \mathbb{R}^m$ and $H \in \mathbb{R}^{m \times n}$ (usually denoted S), the structural matrix that encodes the hierarchical linear constraint.

The target vector $\mathbf{y} \in \mathbb{R}^m$ enjoy some hierarchical structure if it verifies:

$$\mathbf{y} = H\mathbf{y}_{1:n} \quad \text{with for example} \quad H = \begin{bmatrix} \text{Id}_2 \\ 1 & 1 \end{bmatrix} \quad \text{which corresponds to}$$


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graph TD
    y3 --- y1
    y3 --- y2
  
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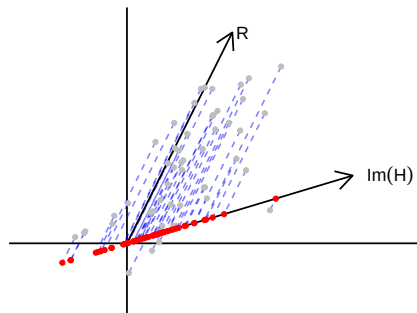
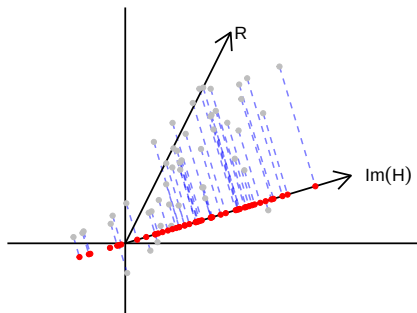
Problem: In general, a forecast $\hat{\mathbf{y}} = \hat{\boldsymbol{\mu}}(\mathbf{x})$ output by a black-box procedure $\mathcal{A}$ does not verify the linear constraint, even if its target $\mathbf{y}$ does.

Hierarchical Data and Forecast Reconciliation

Forecast Reconciliation (Hyndman et al., 2011) consists in obtaining a new forecast at the bottom level $\tilde{\mathbf{y}}_{1:n}$ from a linear transform of $\hat{\mathbf{y}}$ to finally get $\tilde{\mathbf{y}}$ using the structural matrix:

$$\tilde{\mathbf{y}} = H\tilde{\mathbf{y}}_{1:n} \quad \text{with} \quad \tilde{\mathbf{y}}_{1:n} \stackrel{\text{def}}{=} G\hat{\mathbf{y}}$$

Let W be a positive definite matrix, Panagiotelis et al. (2021) studied oblique projections P_W onto $\text{Im}(H)$, which corresponds to $G = (H^\top W H)^{-1} H^\top W$.



Hierarchical Ellipsoidal SCP for joint coverage: Algorithm 3

Split the data between $\mathcal{D}_{\text{train}}$, $\mathcal{D}_{\text{estim}}$ and $\mathcal{D}_{\text{calib}}$

$$\hat{\mu}(\cdot) = \mathcal{A}((\mathbf{x}_t, \mathbf{y}_t)_{t \in \mathcal{D}_{\text{train}}})$$

Compute a symmetric definite positive matrix $A = \mathcal{E}((\mathbf{x}_t, \mathbf{y}_t)_{t \in \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{estim}}})$

Consider the reconciled regressor $\tilde{\mu}(\cdot) = P_A \hat{\mu}(\cdot)$ instead of $\hat{\mu}(\cdot)$

for $t \in \mathcal{D}_{\text{calib}}$, let $\tilde{\mathbf{y}}_t = \tilde{\mu}(\mathbf{x}_t)$ and compute the non-conformity score $\hat{s}_t = \|\mathbf{y}_t - \tilde{\mathbf{y}}_t\|_A$

Set $\hat{E}(\cdot) = \left\{ \mathbf{y} \in \mathbb{R}^m : \|\mathbf{y} - \tilde{\mu}(\cdot)\|_A \leq \hat{s}_{(\lceil (\mathcal{T}_{\text{calib}} + 1)(1 - \alpha) \rceil)} \right\}$ and return $\hat{E}(\mathbf{x}_{\mathcal{T}+1})$

Conformal Prediction for Hierarchical Data: Joint Coverage

Theorem 2 [informal statement]

Algorithms 1 and 3, used with any regression algorithm $\mathcal{A}$ and any estimation procedure $\mathcal{E}$, ensure that whenever data $(\mathbf{x}_t, \mathbf{y}_t)_{1 \leq t \leq T+1}$ is i.i.d.,

$$\mathbb{P}(\mathbf{y}_{T+1} \in \check{E}(\mathbf{x}_{T+1})) \approx 1 - \alpha \quad \text{and} \quad \mathbb{P}(\mathbf{y}_{T+1} \in \dot{E}(\mathbf{x}_{T+1})) \approx 1 - \alpha.$$

In addition, Algorithm 3 outputs prediction ellipsoids $\dot{E}$ that are uniformly more efficient than the prediction ellipsoids $\check{E}$ output by Algorithm 1:

$$\forall \mathbf{x} \in \mathbb{R}^d, \quad \mathfrak{L}_m(\dot{E}(\mathbf{x})) \leq \mathfrak{L}_m(\check{E}(\mathbf{x})).$$

Sketch of proof for the efficiency: we get $\dot{q}_{1-\alpha} \leq \check{q}_{1-\alpha}$ by a Pythagorean theorem and it entails a similar inequality about volumes: $\mathfrak{L}_m(\dot{E}(\mathbf{x})) \leq \mathfrak{L}_m(\check{E}(\mathbf{x}))$.

Hierarchical Component-wise SCP: Algorithm 4

Split the data between $\mathcal{D}_{\text{train}}$, $\mathcal{D}_{\text{estim}}$ and $\mathcal{D}_{\text{calib}}$

$$\hat{\mu}(\cdot) = \mathcal{A}((\mathbf{x}_t, \mathbf{y}_t)_{t \in \mathcal{D}_{\text{train}}})$$

Consider the reconciled regressor $\tilde{\mu}(\cdot) = P\hat{\mu}(\cdot)$ instead of $\hat{\mu}(\cdot)$

for $t \in \mathcal{D}_{\text{calib}}$, let $\tilde{\mathbf{y}}_t = \tilde{\mu}(\mathbf{x}_t)$ and compute the non-conformity score $\tilde{\mathbf{s}}_t = \mathbf{y}_t - \tilde{\mathbf{y}}_t$

for each $i \in [m]$, order the non-conformity scores: $\tilde{s}_{(1),i} \leq \dots \leq \tilde{s}_{(\tau_{\text{calib}}),i}$

Set and return $\tilde{C}_i(\mathbf{x}_{T+1}) = \left[\tilde{\mu}_i(\mathbf{x}_{T+1}) + \tilde{s}_{(\lfloor (\tau_{\text{calib}}+1)\alpha/2 \rfloor),i}, \tilde{\mu}_i(\mathbf{x}_{T+1}) + \tilde{s}_{(\lceil (\tau_{\text{calib}}+1)(1-\alpha/2) \rceil),i} \right]$

Results (Component-wise coverage)

Assumption 1: The non-conformity scores $\hat{\mathbf{s}}_t = \mathbf{y}_t - \hat{\mathbf{y}}_t$, for $t \in \mathcal{D}_{\text{calib}} \cup \{T+1\}$ of the plain multivariate version of SCP (Algorithm 2) are i.i.d. This is in particular the case when data $(\mathbf{x}_t, \mathbf{y}_t)_{1 \leq t \leq T+1}$ is i.i.d.

Theorem 3: Component-wise coverage

Fix $\alpha \in (0, 1)$. Algorithm 4, used with any regression algorithm $\mathcal{A}$ and any matrix P such that $PH = H$, ensures that whenever Assumption 1 (i.i.d. scores) holds,

$$\forall i \in [m], \quad \mathbb{P}(y_{T+1,i} \in \tilde{C}_i(\mathbf{x}_{T+1})) \geq 1 - \alpha.$$

In addition, if the non-conformity scores $(\hat{\mathbf{s}}_t)_{t \in \mathcal{D}_{\text{calib}} \cup \{T+1\}}$ are almost surely distinct, then

$$\forall i \in [m], \quad \mathbb{P}(y_{T+1,i} \in \tilde{C}_i(\mathbf{x}_{T+1})) \leq 1 - \alpha + \frac{2}{T_{\text{calib}} + 1}.$$

Results (Small-length – weak version)

Assumption 2: The (i.i.d.) non-conformity scores $\hat{\mathbf{s}}_t = \mathbf{y}_t - \hat{\mathbf{y}}_t$, for $t \in \mathcal{D}_{\text{calib}}$, of the plain multivariate version of SCP (Algorithm 2) follow some elliptical distribution (whose parameters are unknown to the learner).

Theorem 4: Small-length – weak version

Let $\mathbf{w} \in \mathbb{R}^m$ be a vector of positive numbers. Under Assumptions 1 and 2 (i.i.d. scores with elliptical distribution), Algorithm 4 run with $P = P_{\mathbf{w}}$, where

$$P_{\mathbf{w}} \stackrel{\text{def}}{=} H(H^\top \text{diag}(\mathbf{w})H)^{-1}H^\top \text{diag}(\mathbf{w}), \quad (1)$$

provides more efficient prediction rectangles than Algorithm 2 in the following sense:

$$\mathbb{E} \left[\sum_{i=1}^m w_i \mathcal{L}_1(\tilde{C}_i(\mathbf{x}_{T+1}))^2 \right] \leq \mathbb{E} \left[\sum_{i=1}^m w_i \mathcal{L}_1(\hat{C}_i(\mathbf{x}_{T+1}))^2 \right].$$

Results (Small-length – strong version)

Assumption 3: The (i.i.d.) non-conformity scores $\hat{\mathbf{s}}_t = \mathbf{y}_t - \hat{\mathbf{y}}_t$, for $t \in \mathcal{D}_{\text{calib}}$, of the plain multivariate version of SCP (Algorithm 2) have a bounded second-order moment. We denote by Σ their (positive definite) covariance matrix.

Theorem 4: Small-length – strong version

Under Assumptions 1, 2 and 3, Algorithm 4 run with $P = P_{\Sigma^{-1}}$, where

$$P_{\Sigma^{-1}} \stackrel{\text{def}}{=} H(H^\top \Sigma^{-1} H)^{-1} H^\top \Sigma^{-1}, \quad (2)$$

provides more efficient prediction rectangles than with any other choice of a projection matrix onto $\text{Im}(H)$:

$$\forall i \in [m], \quad \mathbb{E} \left[\mathfrak{L}_1(\tilde{C}_i^*(\mathbf{x}_{T+1}))^2 \right] \leq \mathbb{E} \left[\mathfrak{L}_1(\tilde{C}_i(\mathbf{x}_{T+1}))^2 \right]$$

and in particular

$$\forall i \in [m], \quad \mathbb{E} \left[\mathfrak{L}_1(\tilde{C}_i^*(\mathbf{x}_{T+1}))^2 \right] \leq \mathbb{E} \left[\mathfrak{L}_1(\hat{C}_i(\mathbf{x}_{T+1}))^2 \right].$$

Discussion and Future works

Discussion.

- The Elliptical assumption is not distribution-free
- The optimality result of Theorem 4 is not robust

Future works.

- Extension to Time Series
- Extension to Hierarchies that vary over time
- Use case of EV for smart charging (our approach has already been successfully implemented on daily energy demand in Palo Alto (Principato et al., 2024))

References

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- Panagiotelis, A., Athanasopoulos, G., Gamakumara, P., and Hyndman, R. J. (2021). Forecast reconciliation: A geometric view with new insights on bias correction. *International Journal of Forecasting*, 37(1):343–359.
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The projection matrices considered in practice

The projection matrix can be constant (OLS) or data-based (WLS, MinT and Combi).

Data-based forecast reconciliation requires an additional dataset $\mathcal{D}_{\text{estim}}$ of cardinality T_{calib} on which we get $P = \mathcal{P}(\hat{\mu}, (\mathbf{x}_t, \mathbf{y}_t)_{t \in \mathcal{D}_{\text{estim}}})$. Typically, it corresponds to $P = \mathcal{P}'(\hat{\Sigma})$, with $\hat{\Sigma}$ being the empirical estimate of the covariance matrix.

Nickname	Parameter	Expression
OLS	P_1	$H(H^\top H)^{-1}H^\top$
WLS	$\mathcal{P}_{\text{WLS}}(\hat{\Sigma})$	$H(H^\top \text{Diag}(\hat{\Sigma})^{-1}H)^{-1}H^\top \text{Diag}(\hat{\Sigma})^{-1}$
MinT	$\mathcal{P}_{\text{MinT}}(\hat{\Sigma})$	$H(H^\top \hat{\Sigma}^{-1}H)^{-1}H^\top \hat{\Sigma}^{-1}$
Combi	$\mathcal{P}_{\text{Combi}}(\hat{\Sigma})$	$\frac{1}{3}(P_1 + \mathcal{P}_{\text{WLS}}(\hat{\Sigma}) + \mathcal{P}_{\text{MinT}}(\hat{\Sigma}))$

Here, $\hat{\Sigma}$ can be nearly singular so we consider its Moore-Penrose pseudo-inverse.

Key ideas for the proofs of Theorems 3–4 [informal]

$\tilde{\mathbf{s}}_t - \mathbb{E}[\tilde{\mathbf{s}}_t]$ are i.i.d. according to a centered elliptical distribution as $t \in \mathcal{D}_{\text{calib}}$ so their i -th components have the same distribution ν up to scaling factors denoted by $\sqrt{\gamma_i}$. Thus,

$$\mathbb{E} \left[\sum_{i=1}^m w_i \mathfrak{L}_1(\tilde{C}_i(\mathbf{x}_{T+1}))^2 \right] = \mathbb{E}[L_\alpha^2] \sum_{i \in [m]} w_i \gamma_i$$

with $L_\alpha = \nu_{(\lceil (T_{\text{calib}}+1)(1-\frac{\alpha}{2}) \rceil)} - \nu_{(\lfloor (T_{\text{calib}}+1)\frac{\alpha}{2} \rfloor)}$ and $(\nu_t)_{t \in \mathcal{D}_{\text{calib}}} \stackrel{i.i.d.}{\sim} \nu$.

γ_i is the (i, i) -th element of a matrix of the form $P_{\mathbf{w}} \Gamma P_{\mathbf{w}}^\top$, where Γ is symmetric positive semi-definite. A similar result holds for the $\hat{C}_i$, with scaling factors given by the diagonal elements of Γ . It thus suffices to show that

$$\sum_{i \in [m]} w_i \Gamma_{i,i} = \text{Tr}(\text{diag}(\mathbf{w}) \Gamma) \geq \text{Tr}(\text{diag}(\mathbf{w}) P_{\mathbf{w}} \Gamma P_{\mathbf{w}}^\top) = \sum_{i \in [m]} w_i (P_{\mathbf{w}} \Gamma P_{\mathbf{w}}^\top)_{i,i}.$$